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Introduction to Waveform Generation Applications in Ultrasound

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Introduction

Medical ultrasound imaging has undergone many improvements since its initial development, and clinical systems produce very sophisticated 2D images in B-mode, Doppler, and 3-Dimensional volume renderings.

Many ideas for improving image quality and information content have been proposed over the years, but have languished in academic and industrial labs for lack of practical tools with which to implement the concepts.

With the advent of “open” software-based ultrasound systems that provide developer with access to raw received data, easily programmable control of the acquisition sequence and beam parameters, as well as access to all of the processing functions in the imaging chain, a new era of innovation is underway. The software-based ultrasound system has led to the development of fundamentally new approaches to ultrasound imaging, including elastographic methods, new beamforming variants, and the exploration of the use of coded excitations. Applications for coded transmissions include extending the useful bandwidth of a transducer, extending the imaging depth in attenuating media, extending the frame rate (simultaneous acquisitions using independent transmissions), and time-reversed waveforms.

Conventional transmit waveforms

Conventional transmit waveforms for B-mode imaging are typically short pulses that offer the best pulse-echo resolution given the center frequency of the transducer’s passband. Generally, these pulses can be said to have small time-bandwidth product, because they are only slightly longer than the transducer’s own impulse response. Ultrasound echoes using such pulses are typically used directly in beamforming and image reconstruction; in that case, axial resolution is directly related to pulse length. For these pulses, the maximum signal strength is achieved at maximum allowable drive voltage, which is limited by either system capability, transducer technology, or regulatory safety considerations. To prevent formation of bubbles due to acoustic cavitation, safety limits on the instantaneous rarefactional pressure (via the Mechanical Index - MI) place a maximum cap on drive voltage for a given transmit beam, thus limiting the signal strength and the maximum imaging depth. The average power carried by the acoustic field is limited by thermal safety considerations (via the Thermal Index - TI). Details regarding the parameters on which the MI and TI depend are beyond the scope of this paper, but it is important to recognize that regulatory limits often determine the practical maximum depth of penetration.

Coded excitations

Coded excitations are transmit waveforms that are substantially longer than the transducer's impulse response (large time-bandwidth product) and usually require some form of additional signal processing to time-compress the pulses before high resolution images can be produced by conventional reconstruction techniques. At a given drive voltage, long time- bandwidth pulses contain substantially more total energy than conventional pulses. After pulse compression, echo signals appear to have been produced using short pulses at voltages far greater than the actual drive voltage. Thus, in their most basic application, coded transmits can be used to extend the maximum imaging depth when it is limited by the MI.

Another common application of coded transmissions is to increase the acquisition frame rate by performing normally sequential acquisitions simultaneously. If different coded waveforms can be transmitted simultaneously on separate array channels, pulse encoding can be done in such a way as to make the individual channel signals distinguishable in post- processing. Thus, many different T-R experiments may be conducted in the same acquisition. For example, several different angle plane wave transmissions can be encoded in one set of transmit pulses, and the echoes can be separated in post-processing and reconstructed independently as though they were acquired in sequential events. The length of the transmit pulses depends on the permissible degree of remaining "crosstalk" between separate plane wave angle experiments, and the bandwidth of the transducer elements.

Time-reversal experiments also require use of long time-bandwidth waveforms, because the waveforms result from measurements of highly reverberant fields, resulting from extensive multiple scattering. These waveforms are naturally pulse compressed by propagating back through the reverberant medium and can achieve enormous focusing and compression gains.

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